AIAA SciTech Forum
12-16 January 2026, Orlando, FL
AIAA SCITECH 2026 Forum

10.2514/6.2026-2328

Cross-Stream PIV Characterization of the Prandtl-D3C Wake

Charles Cain*
University of Dayton, Dayton, Ohio, 45469, United States

Jessica DeMoor†
University of Dayton, Dayton, Ohio, 45469, United States

Sidaard Gunasekaran‡
University of Dayton, Dayton, Ohio, 45469, United States

This paper presents the cross-stream particle image velocimetry (PIV) measurements of the wake structure behind a Prandtl-D3C wing configuration tested in the University of Dayton Low-Speed Wind Tunnel. Previous computational and experimental efforts have suggested that bell-shaped lift distributions may suppress or delay tip vortex roll-up. However, direct experimental validation using cross-plane PIV has been lacking. In this study, PIV data were acquired at three downstream planes (2c, 3c, and 4c) under varying angles of attack and compared against FlightStream® panel method simulations. Results show a distinct inboard vorticity structure with no evidence of conventional wingtip vortex near the design condition, providing strong support for the hypothesized wake behavior. Background subtraction and multi-plane stitching were employed to resolve weak cross-stream velocity components. These findings contribute new experimental insight into the wake dynamics of non-elliptically loaded wings and support ongoing interest in low-vorticity, high-efficiency lifting surface designs.

Nomenclature

$\bar{c}$	=	Mean Aerodynamic Chord
c	=	Local chord
b	=	Wingspan
U, V, W	=	Velocity in X, Y, Z Directions
ω_x	=	X-Vorticity
Γ	=	Circulation
Γ^*	=	Normalized Circulation
α_{local}	=	Local angle of attack (deg)
α_{Wing}	=	Wing angle of attack based on root chord (deg)
C_L	=	Coefficient of Lift
C_D	=	Coefficient of Drag

I. Introduction

The Prandtl-D wing concept, with its associated bell-shaped lift distribution, has attracted growing interest in recent years due to its potential for induced drag reduction and unconventional wake characteristics. Originally introduced by Ludwig Prandtl in 1933 [1], the bell-shaped lift distribution deviated from the conventional elliptical loading, shifting the focus from fixed-span designs to those constrained by bending moments. Prandtl showed that, for a given structural constraint, a non-elliptical distribution could reduce induced drag more effectively than the elliptical distribution, leading to a more efficient overall aerodynamic configuration [1]. Hunsaker and Phillips [2] revisited and translated this seminal work, emphasizing that the bell-shaped span load produces a more gradual

* Graduate student, Department of Mechanical and Aerospace Engineering, AIAA Member. cainc3@udayton.edu
† Undergraduate student, Department of Mechanical and Aerospace Engineering, AIAA Member. demoorj1@udayton.edu
‡ Associate Professor, Department of Mechanical and Aerospace Engineering, AIAA Senior Member. gunasekarans1@udayton.edu

tapering of lift near the tips and diminished tip vortices. A modern resurgence of interest in the Prandtl-D concept has been driven in part by experimental work conducted at NASA Armstrong Flight Research Center. Bowers et al. [3] designed and flew a series of Prandtl-D aircraft that demonstrated coordinated turns without rudders, a phenomenon termed "proverse yaw." These efforts culminated in the PRANDTL-M concept, which explored the feasibility of Mars atmospheric flight using the bell-shaped loading approach [4]. A NASA technology success report [5] summarized that the Prandtl-D configuration enabled up to 12% drag reduction and laid the groundwork for rudderless flight control via wing twist [6]. Numerical and experimental studies have since sought to validate the performance advantages and flow mechanisms associated with this novel span loading. Ranjan and Ansell [7] conducted a RANS-based comparative study between elliptically and bell-loaded wings, revealing delayed vortex roll-up and a wake structure with minimal near-field vorticity in the bell-loaded case. Similarly, Bragado-Aldana et al. [8] and Hoeijmakers et al. [9] provided analytic and computational support for the claim that non-elliptical lift distributions shift the center of vorticity inboard, minimizing wake curvature and improving span efficiency. Phillips et al. [10] derived optimal lift distributions for minimum induced drag under varying structural constraints, reinforcing that elliptical loading is not universally optimal.

In recent CFD investigations, Hammer and Garmann [11], [12] simulated Prandtl-D and related bell-loaded configurations under various flow conditions and geometries. They employed a multi-fidelity simulation framework that combined vortex-lattice methods (VLM), Euler simulations, and Reynolds-averaged Navier–Stokes (RANS) computations to characterize the flow physics and performance of bell-shaped span-loaded wings. At design conditions near α = 8°, the flow fields exhibited nearly planar wakes with reduced or absent tip vortices, instead forming distributed vorticity sheets. High-fidelity RANS simulations showed that the suppression of traditional tip-vortex structures persisted even at moderate Reynolds numbers, with weak separation localized near the trailing edge. Figure 1 reproduces a representative result from that study, highlighting the near-planar wake and modified vortex topology. A recent large-eddy simulation (LES) study further extends and corroborates these findings [13].

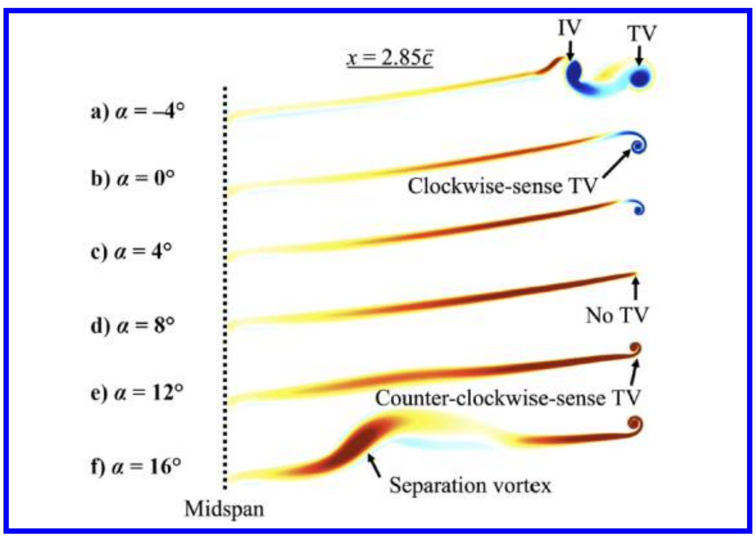

Figure 1. Prandtl-D wake vorticity contours, Hammer and Garman RANS CFD study [12]

Zelenka et al. [14] performed wind-tunnel experiments on a scaled Prandtl-D configuration to investigate its qualitative and quantitative aerodynamic behavior. The study combined surface oil-flow visualization, force-balance measurements, and streamer-based flow tracing, and showed that premature flow separation could occur in models with sharper-than-intended leading edges. More recent experimental work has strengthened the empirical basis for Prandtl-D wake behavior. Pabon et al. [15] and Grace et al. [16] reported force measurements and streamwise PIV data that demonstrated a clear absence of tip-vortex roll-up in the near wake of the Prandtl-D3C and of a non-elliptically loaded wing model, respectively. Cain and Gunasekaran [17] observed similar near-wake characteristics for a rectangular wing with a bell-shaped lift distribution. Additional Prandtl-D experimental results are in ref [26].

In these studies, wake centerline curvature and vortex-topology measurements confirmed that the primary wing vortex was weaker and located further inboard than for comparable elliptically loaded wings.

Figure 2 shows the variation of C_L, C_D, and L/D with angle of attack for the scaled Prandtl-D wing from multiple investigations, including RANS simulations by Hammer and Garmann, panel-method (FlightStream®) predictions, and wind-tunnel measurements [11], [15]. The results show consistent trends across methods. At the design condition near $\alpha \approx 8°$ and $C_L \approx 0.6$, L/D is at its maximum and the computed wakes in both the RANS and panel-method solutions exhibit no coherent tip vortex in the near field. This agreement suggests a strong link between the bell-shaped span loading, the inboard-shifted and weakened vortex system, and the near-planar wake topology.

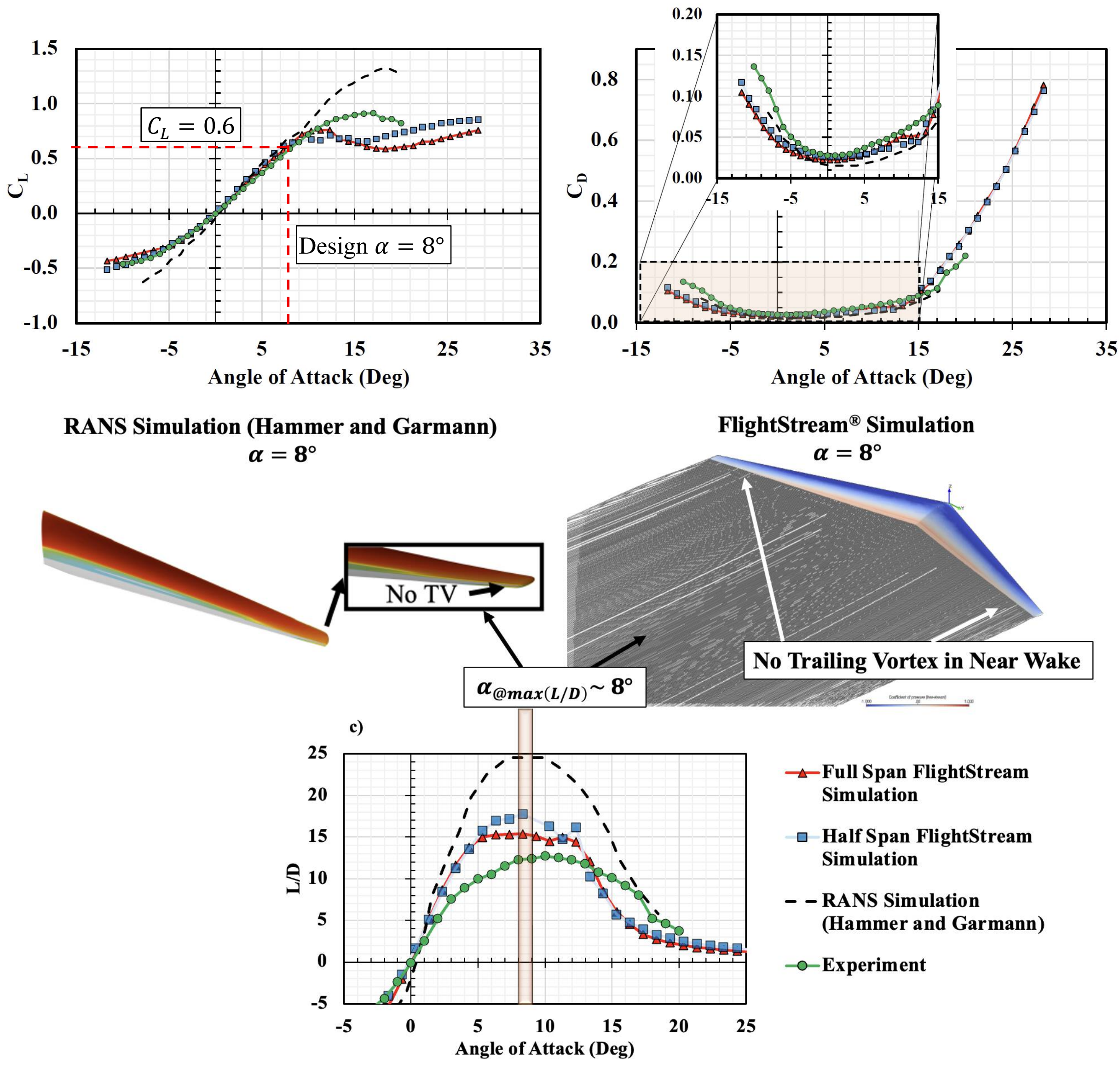

Figure 2. Prandtl-D C_L and C_D data, Pabon, and Hammer and Garman RANS CFD study [15]

Despite these advances, direct cross-stream PIV measurements of the Prandtl-D wake remain scarce. Most prior experimental studies have relied on streamwise-plane PIV or qualitative flow-visualization techniques, which cannot fully resolve the three-dimensional wake roll-up and vortex topology. The present work addresses this gap by presenting, to the best of authors' knowledge, the first systematic cross-stream PIV measurements of the Prandtl-D3C wake over a range of test conditions. These measurements are essential for clarifying whether tip-vortex roll-up is truly suppressed or instead displaced and delayed in bell-span-loaded configurations.

II. Experimental setup

A. University of Dayton Low Speed Wind Tunnel

All experiments were performed in the University of Dayton Low Speed Wind Tunnel (UD-LSWT) using its open-jet configuration. The tunnel features a 16:1 contraction ratio, six anti-turbulence screens, and a square open-jet test section measuring 0.762 m × 0.762 m. The effective test section length is 0.183 m, and a 1.128 m × 1.128 m collector captures the expanded flow and directs it into the diffuser. Freestream velocity was measured using a pitot-static probe connected to a TSI 5825 manometer. A schematic of the wind tunnel setup and relative locations of the wing and cross-stream PIV image locations is shown in Figure 3. Cross-stream PIV images were acquired at three downstream locations: 2c, 3c, and 4c.

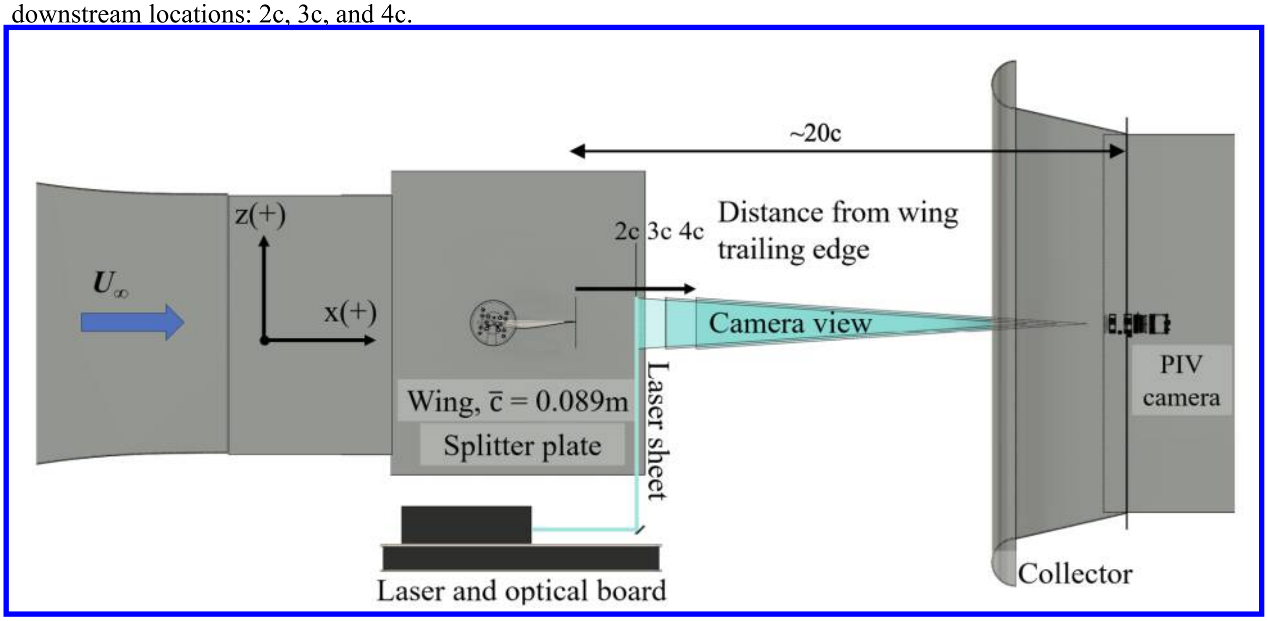

Figure 3 Schematic of UD-LSWT with cross-stream PIV setup and the planes of interrogation at different downstream distances.

Table 1 summarizes the PIV image locations and test conditions for the results captured in this study. The angles of attack were aligned with prior CFD studies with additional points near the 8 deg design angle of attack to bracket that location in case of wind tunnel angularity or accuracy of the AOA setting on the manual traverse.

Table 1. PIV Test Matrix

Location	Image Center (y,z), [2y/b]	Freestream (m/s)	Reynolds Number	FOV (mm)	α_{root} (deg)
2c	(0.89, .08)	26	150,000	186 x 186	0, 4, 7, 8, 9, 12
3c	(0.89, .08)			183 x 183	
4c	(0.89, .08)			186 x 186	
4c inboard	(0.67, .08)			185 x 185	

B. Cross-stream PIV Setup

The wing model was mounted horizontally to a manual rotary stage, which was secured to a vertical splitter plate as shown in Figure 4a. PIV imaging was in the cross-stream capturing the wake corresponding to the outboard 45% of the half-span in composite PIV images at the 4c location. Imaging was performed using an Imperx B2021M CCD camera equipped with a 100 mm lens and electronically controlled lens focus and aperture. The camera provided a pixel resolution of 2072 × 2072 and operated at up to 7.5 Hz in dual-shutter mode. Both the camera and lens assembly were mounted on a manual traverse positioned 20c downstream of the wing trailing edge (at the tip) and within the tunnel's collector section, as shown in Figure 3 and in Figure 4b.

Illumination for particle image velocimetry (PIV) was provided by a Quantel/Big Sky ND-YAG Twin CFR 300 dual-pulsed laser (Figure 3). The laser beam was shaped into a vertical light sheet using a plano concave lens and mirror. Each laser beam had a measured energy of 206.7 mJ at 532 nm. The lasers were water cooled, powered, and controlled by two cooler and power units. A Quantum Composers timing unit synchronized the laser pulses with the camera's split-frame exposures to capture paired images at precisely controlled intervals. The time delay, Δt, between image pairs ranged from 101 μs to 111 μs, adjusted based on predicted and observed seeding particle velocities. The spatial resolution varied from 11,139 to 11,322 pix\/m depending on the camera position, corresponding to a field of view between 183 mm and 186 mm.

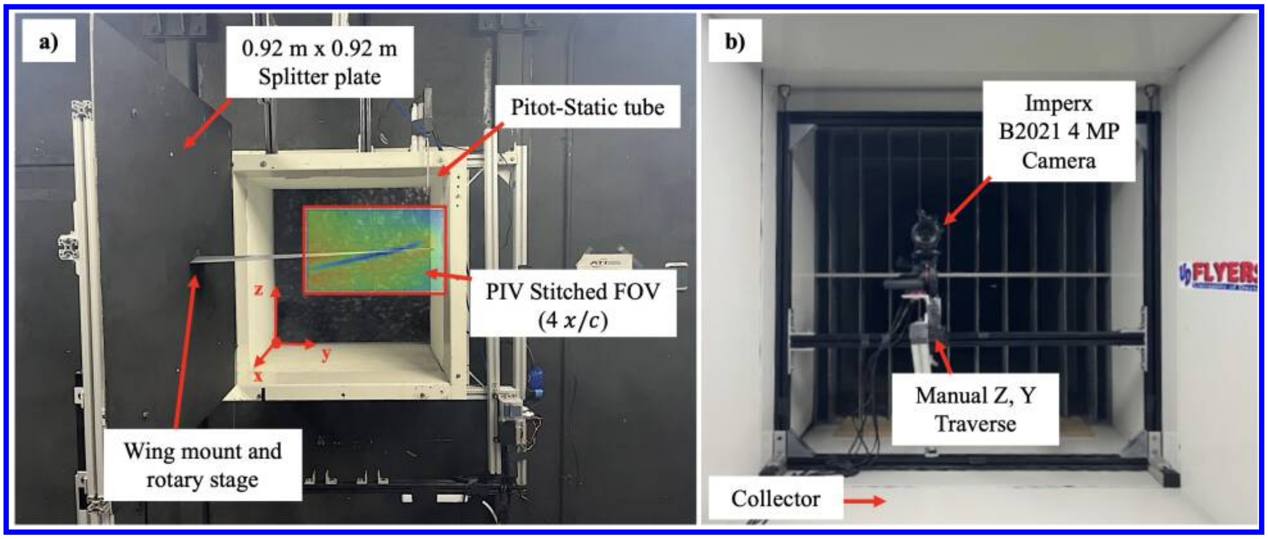

Figure 4 a) Prandtl-D3C half-span wing model mounted on manual rotary stage and splitter plate in open-jet test section b) Imperx B2021M CCD camera (2072 × 2072 pixels) mounted on manual traverse system in collector section for cross-stream wake imaging.

C. Wing Geometry

The Prandtl-D3C model used in this study was 3D-printed, based on geometry specified in [3]. The dimensions of the model and the specifications are shown in Figure 5a. It had a half-span of 0.635m with a mean aerodynamic chord of 0.089m. The wing was swept with a 22° leading edge wing sweep, taper ratio of 0.25, and full-span equivalent geometric aspect ratio of 16.

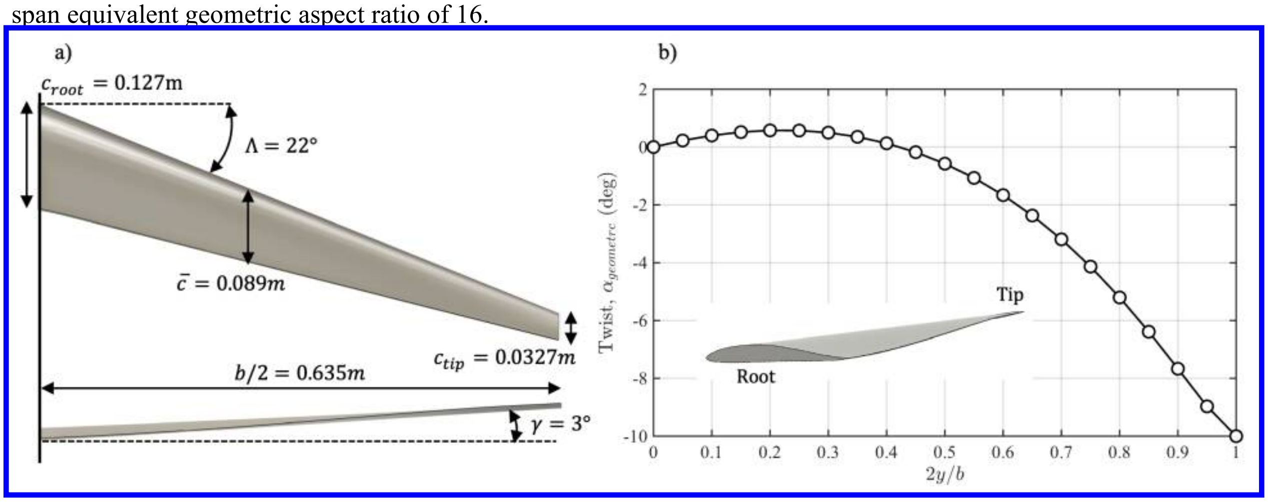

Figure 5 a) Dimensions of Scaled Prandtl-D3C Wing tested b) Half-Span Prandtl-D3C Wing Twist Distribution

This design achieved its lift distribution using a combination of aerodynamic twist, geometric twist, and wing taper. The spanwise wing twist follows that shown in Figure 5b [derived from [3]]. The geometric angle of attack is $0°$ at the wing root and then increases until $2y/b = 0.2$ from which point it decreased to $-10°$. The overall wing angle of attack is referenced at the wing root. A detailed discussion on the theory behind this wing design and how that is applied to create a wing shape is in ref [4].

D. PIV Data Processing

Fluere PIV processing software [18] was used to convert raw image pairs into velocity vector fields. The software employs the algorithm developed by Scarano and Riethmuller [19], and the specific processing parameters are summarized in Table 2. After processing each test point, the resulting vector files were reviewed to identify and remove erroneous or incomplete frames. The remaining valid data, typically drawn from 1,000 valid image pairs out of 2,000 total captures were ensemble-averaged to improve signal quality. Given that the expected cross-stream velocity components were small (generally less than 10% of the freestream velocity), background velocity fields from no-wing baseline tests were acquired and subtracted from the corresponding with-wing datasets to isolate the wake-induced flow. Spurious vectors near the edges of the laser sheet were masked. Mid-span and outboard vector fields were then merged using scattered data interpolation, guided by a manually adjusted image offset parameter. Vorticity fields were computed from the resulting composite velocity maps.

Table 2. Fluere PIV Image Processing Parameters.

Parameter	Pass 1	Pass 2
Interrogation Window (pix)	64 × 64	32 × 32
Window spacing (pix)	32 × 32	16 × 16
Offset	50%	50%
Outlier detection passes	2	
Outlier detection threshold (σ)	2.5	
Window weighting	Gaussian	
Interpolation method	Sinc (7×7)	

E. Numerical Investigation with FlightStream

FlightStream [20] was used to numerically investigate the same models tested in the wind tunnel to obtain the surface pressure distribution on the PRANDTL-D3C wing. FlightStream includes two main types of potential flow solvers: a pressure-based solver and a vorticity solver. The pressure-based solver, which utilizes the pressure fields on the model geometry to determine the aerodynamic loads, is used in the present study. A combination of source and sink panels, along with doublet panels on the trailing edge, is employed to provide a solution for the pressure-based potential flow solver. The pressure-based solver was used in the case to align based on its good agreement with the wind tunnel experimental results in previous studies [21–24]. The boundary layer that the panel code models was assumed to be fully turbulent, with viscous coupling enabled. An additional feature of FlightStream is that it visualizes and tracks trailing vortex filaments, which are allowed to propagate according to the velocities induced by the panel-based solution and the vortex filament system. It uses a relaxed vortex-wake filament method combined with the Lamb–Oseen vortex core model to characterize this trailing vortex filament behavior. The surface mesh of the wing and the splitter plate in FlightStream is shown in Figure 6. The wing platform has a structured mesh with 120 node points spanwise and 80 node points chordwise on both the upper and lower surfaces. The chordwise mesh also has a dual-side growth rate of 1.05 to better resolve the leading and trailing edge geometry. Additionally, a splitter plate of the same size as the one used in the wind tunnel experiment is included in the model to better represent the actual experimental conditions. The splitter plate has 60 node points from side to side and 120 node points downstream.

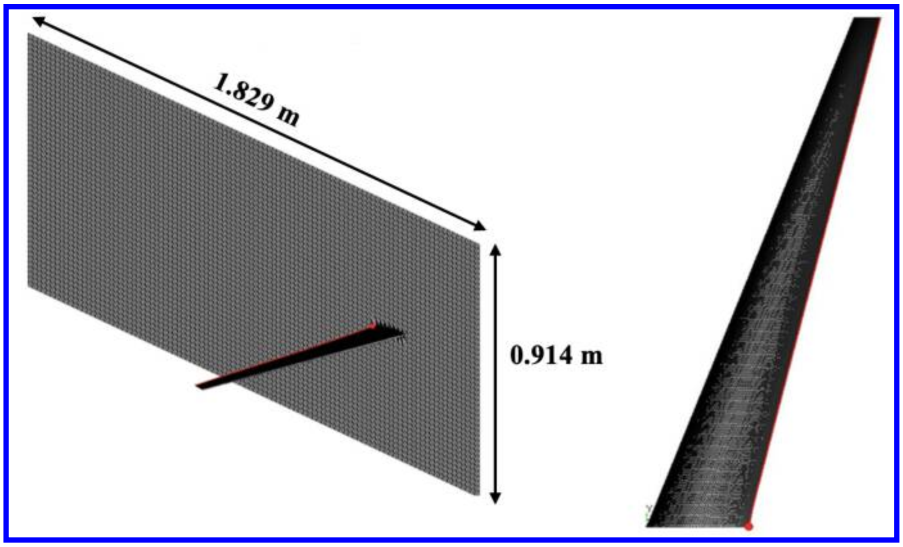

Figure 6 Surface Mesh in FlightStream for a) Isometric View of the Wing and Splitter Plate and b) Top View of the Wing

III. Results

A. Cross-Stream Wake Characteristics at 4c

Cross-stream PIV measurements were acquired at three downstream locations (2c, 3c, and 4c) and six angles of attack (0°, 4°, 7°, 8°, 9°, 12°) to characterize the wake structure of the Prandtl-D3C wing. The angles of attack were selected to show a range of flow qualities in the wake shear layer and wingtip. The analysis focuses primarily on the 4c plane, where wake features are fully developed. The results were similar at 2c and 3c. A key result is the change in the wingtip vortex and inboard pancake vortex behavior for angles of attack below or above the 8° design angle of attack. The wingtip vortex behavior aligns well with the results found in ref [11] and [12]. These highlight the unique nature of this lift distribution and the Prandtl-D wing and align with modeling results.

Figure 7 presents inboard and outboard PIV data stitched together to form one set of vectors data at the 4c downstream plane for angles of attack ranging from 0° to 12°. Each row displays three quantities: normalized cross-stream velocity magnitude with velocity vectors, vertical velocity component showing upwash (red) and downwash (blue), and normalized streamwise vorticity ω_x^*. The integrated non-dimensional circulation is displayed for each angle of attack condition and is calculated numerically from the vorticity present in the corresponding frame. It is consistent with increasing lift with increasing angle of attack, even in the absence of a conventional wingtip vortex. In that case, the circulation is present in the wake shear layer.

At low angles of attack ($\alpha = 0.0°$ and $4.0°$), the wake exhibits characteristics consistent with a conventional wing behavior with the wingtip at a negative angle of attack. At $\alpha = 0.0°$, a strong counter-rotating tip vortex is clearly visible near the wingtip region at $2y/b \approx 0.95$. The velocity magnitude plots show organized recirculation patterns with concentrated vorticity at the vortex core. Strong vortex structures are evident in the vorticity contours, with peak normalized vorticity values reaching $\omega^* \approx 0.8 - 1.0$. Consistent with this vorticity, the vertical velocity contours reveal distinct upwash regions (red, $W/U_\infty > 0$) in the center of the measurement domain with downwash (blue) near the wingtip. The cross-stream velocity vectors form tightly organized spiral patterns indicative of strong clockwise (CW) vortex roll-up, with maximum velocity magnitudes approaching 10% of freestream near the concentrated vortex. Also present in this flow is an inboard area of circulation characterized by a distinctive inboard "pancake" vortex that is coincident with the wake shear layer. This moves outboard as the angle of attack increases until it merges with a conventional wingtip vortex at higher angles of attack. At the design angle of attack this pancake vortex is the main source of circulation in the data frame.

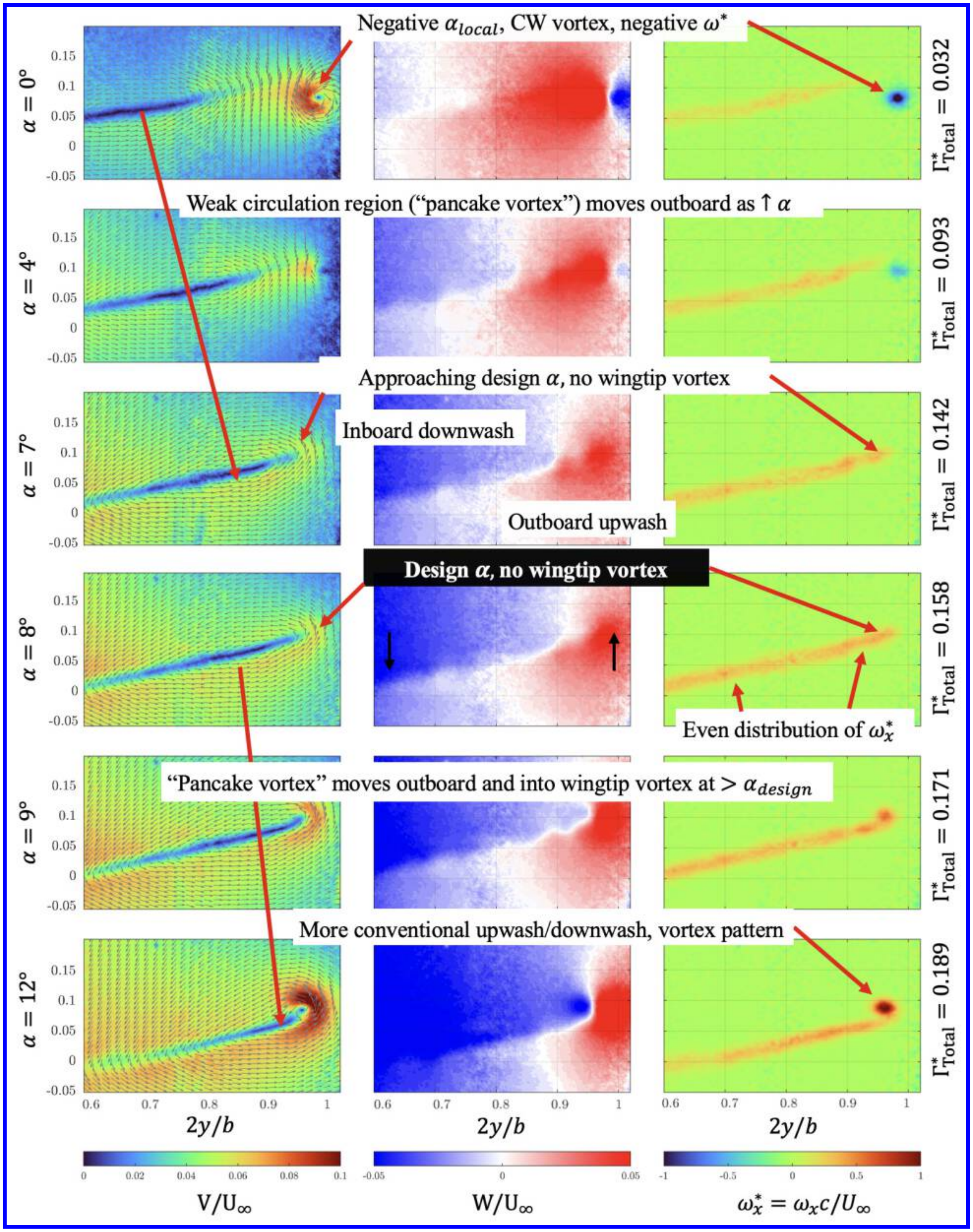

Figure 7 Cross-stream PIV measurements at 4c downstream for $\alpha = 0.0°$ to $\alpha = 12°$

At $\alpha = 4.0°$, a conventional wingtip vortex structure remains, but is decreasing in strength, with the concentrated vortex located near $2y/b \approx 0.90 - 0.93$. The vorticity magnitude remains substantial with $\omega^* \approx 0.5$, and the velocity vectors continue to show some organized CW spiral motion characteristic of coherent vortex structures. The vertical velocity field shows strong downwash with gradients in both the pancake vortex and residual wingtip vortex locations. The integrated circulation nearly triples from α = 0° to 4°, reflecting the increased lift generation. At $\alpha = 7.0°$, a transformation in wake topology becomes obvious. The traditional tip vortex structure disappears and the vorticity in the wake is confined to the shear layer and is distributed evenly. The cross-stream velocity vectors show diminishing organized rotation near the geometric wingtip. The vertical velocity contours reveal a transition zone where the sharp downwash-upwash interface begins to smooth out. This represents the onset of the unique wake behavior associated with bell-shaped lift distributions approaching their design condition. The weakening and inboard migration of vorticity at $\alpha = 7.0°$ serves as a clear precursor to the complete vortex suppression observed at the design angle of attack.

At $\alpha = 8.0°$, no discernible conventional wingtip vortex is observed. Instead of a concentrated vortex at the wingtip, the wake is now completely characterized by a distinctive inboard "pancake" vortex which is a flattened region of weak, distributed circulation between $2y/b = 0.75 - 0.78$ that coincides spatially with the wake shear layer. The vorticity contours at $\alpha = 8.0°$ show a broad and a uniform region of positive streamwise vorticity with peak values of $\omega_x^* \approx 0.5$ with no concentrated vortex structure evident. The nature of this structure is fundamentally different from conventional vortex structures, which typically exhibit sharp gradients and concentrated zone rotation at the wingtip. The cross-stream velocity magnitude at $\alpha = 8.0°$ remains uniformly low throughout the measurement domain, with peak values below 6-7% of freestream and a smooth transition from downwash to upwash occurring near $2y/b \approx 0.70 - 0.75$, with no sharp gradient characteristic of concentrated vortex structures. This gradual transition, with vertical velocity magnitudes remaining below $W/U_\infty \approx 0.03 - 0.04$, is consistent with theoretical predictions for this lift distributions. The circulation value indicates positive lift generation despite the complete absence of traditional tip vortices.

At $\alpha = 9.0°$, the wake structure begins transitioning back toward more conventional behavior. A counter-clockwise rotating tip vortex starts to form with vorticity magnitude increasing to $\omega_x^* \approx 0.6 - 0.7$. The inboard pancake vortex structure persists as a secondary feature visible near $2y/b \approx 0.72 - 0.88$, creating a bimodal vorticity distribution. The velocity vectors begin showing organized spiral patterns near the wingtip, and the vertical velocity field exhibits increasing downwash-upwash gradients. This marks a return to more conventional wake topology as the wing operates beyond its design circulation distribution.

By $\alpha = 12°$, strong conventional wingtip vortices dominate the wake, with highly concentrated vorticity structures ($\omega_x^* \approx 0.9 - 1.0$, dark red/brown regions) located near $2y/b = 0.92 - 0.95$. Organized spiraling flow patterns are clearly visible in both velocity vectors and vorticity contours. The cross-stream velocity magnitude increases dramatically, with peak values exceeding 10% of freestream near the vortex center. The vertical velocity field shows strong, localized regions with sharp gradients at the vortex core locations, characteristic of conventional high-lift wing wakes. The integrated circulation is the highest among all test conditions, reflecting the increased lift coefficient at this angle of attack. A remnant of the inboard pancake vortex exists, but as it is pulled toward the wingtip vortex, the wake structure is nearly indistinguishable from that of a conventional wing operating at high lift.

B. Cross-Stream Wake Characteristics at Different x/c

As mentioned earlier, cross-stream PIV measurements were also acquired at $x/c = 2$ and $x/c = 3$ for all angles of attack tested as shown in Table 1. The cross-stream wake characteristics were compared at the different downstream distances for three representative angles of attack: $\alpha = 0°$, 8° (design condition), and 12° in Figure 8. The contours in Figure 8 show the corresponding normalized streamwise vorticity fields, ω_x^*. At $\alpha = 0°$, the wake topology is consistent with that of a lightly loaded conventional wing. A compact, clockwise-sense tip vortex is clearly visible near the outboard edge of the field of view at all three downstream planes. As x/c increases from 2 to 4, the vortex core remains approximately aligned with the geometric wingtip, but its peak ω_x^* decreases and the vorticity distribution broaden, indicating gradual viscous diffusion and entrainment. The overall wake centerline remains nearly straight, with limited curvature or inboard migration.

In contrast, at the design condition of $\alpha = 8°$, no conventional wingtip vortex can be identified at any of the three downstream planes. Instead, the wake is dominated by a diffuse inboard vorticity sheet that coincides with the "pancake" vortex described in the 4c results in Figure 7. As the flow convects from $2c$ to $4c$, the wake centerline exhibits a pronounced curling motion. The high ω_x^* region tilts and shifts inboard, while the vorticity distribution remains broad and relatively low in magnitude. This progressive inboard bending of the wake, combined with the persistent absence of a concentrated tip vortex, is a key signature of the bell-shaped span loading and directly supports the tip vortex free wake topology predicted by prior CFD and panel-method studies.

At $\alpha = 12°$, the wake again resembles that of a conventional high-lift wing. A strong, counter-clockwise-sense tip vortex is present at $x/c = 2$, with large positive ω_x^* concentrated in a compact core near the tip. With increasing downstream distance, the vortex remains coherent, but its peak vorticity decreases and the core expands, consistent with viscous diffusion. The inboard "pancake" structure is still visible but becomes increasingly subordinate to the dominant tip vortex. By $x/c = 4$, the wake topology is effectively that of a conventional wing with a strong trailing vortex system. Figure 8 shows that the Prandtl-D3C wake evolution is strongly dependent on the angle of attack and similar wake characteristics are seen in all downstream distances tested. Only near the design condition does the wake remain tip-vortex-free across all downstream stations, with circulation carried primarily by a weak, inboard vorticity sheet that curls gently as it convects downstream. This behavior contrasts sharply with the persistent tip vortices observed at $\alpha = 0°$ and $\alpha = 12°$, and provides further experimental confirmation that the bell-shaped lift distribution fundamentally alters wake roll-up and tip vortex topology.

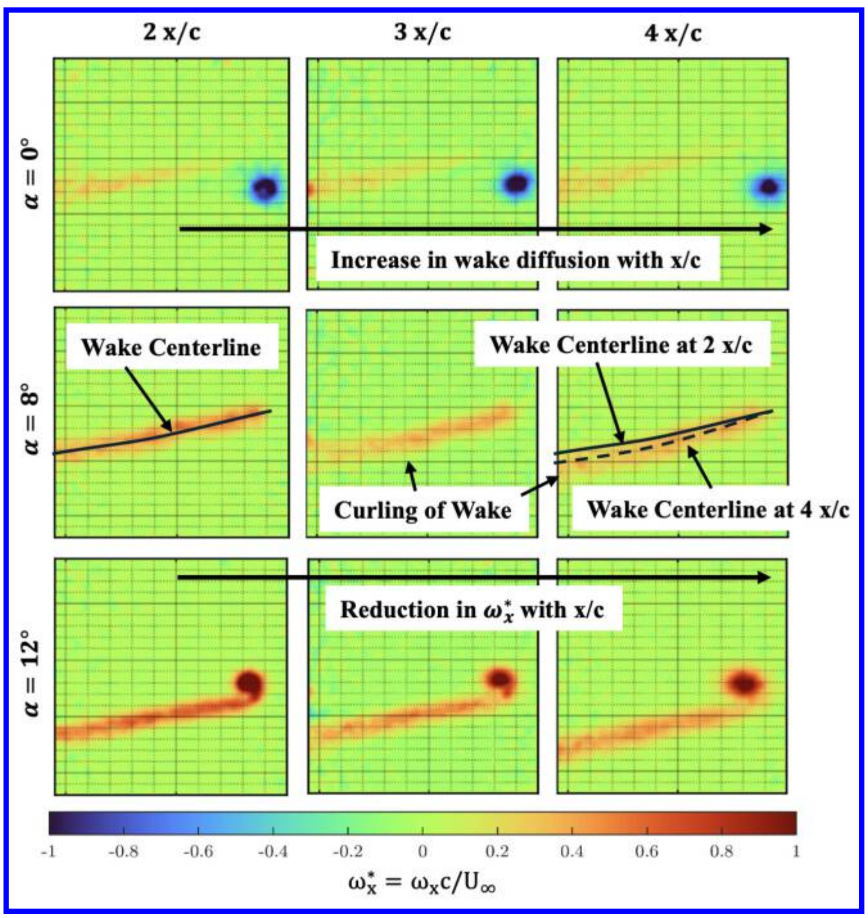

Figure 8 Cross-stream evolution of normalized streamwise vorticity for Prandtl-D3C at α = 0°, 8°, 12°, showing design-condition tip-vortex-free, inboard-curled wake.

C. FlightStream Results

The cross-stream wake evolution behind the Prandtl-D3C wing simulated in the panel method software, FlightStream®, over a range of angles of attack that from 0° to 12°. In addition to calculating aerodynamic forces based on the panel method solution method (see Figure 2), FlightStream® also tracks trailing vortex filaments, which are allowed to move according to the velocities induced by the panel strengths and the vortex filament system itself. Figure 9 shown the geometric solution of this wake behavior at the design α = 8°. The trailing vortex filaments are colored by downstream distance and extend 50 chord lengths (50c) behind the wingtip trailing edge. This result agrees with the PIV data shown in Figure 8, and even in the far-field, shows an inboard curved non-planar wake consistent with the PIV velocity and vorticity profiles. Remarkably, there is not even one rotation of the wingtip vortex filaments by 50c. The only conventional-looking vortex present is from the wing root / splitter plate intersection, which is simulated to mimic the PIV experiment setup.

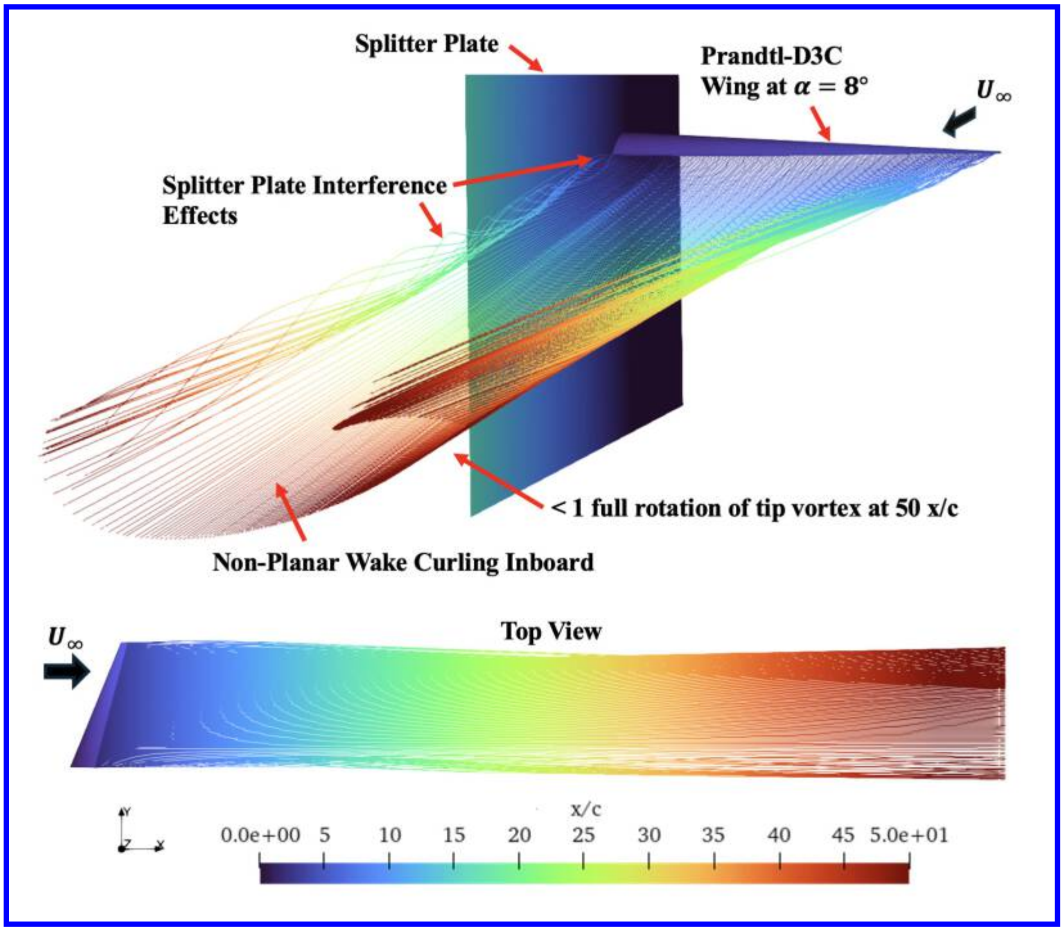

Figure 9. FlightStream Prandtl-D Vortex Filaments, $\alpha = 8°$ to $x/c = 50$

Figure 10 shows an aft view of the wing with the panel method trailing vortex filaments and illustrates how the wake topology changes with angle of attack. At $\alpha = 0°$ a clockwise wingtip vortex roll up is evident, with a strong coherent spiral near the tip. In addition, a counter-clockwise pancake or wing vortex appears near the mid semispan, indicated by the folded vortex filaments. The coexistence of these two vortices matches the PIV observations at the same angle of attack in Figure 7. As α increases, the tip vortex structure weakens and the pancake vortex migrates outboard. At the design condition, $\alpha = 8°$, no distinct tip vortex is present. The wake is instead dominated by weak inboard circulation associated with the folded filaments. In the experiment this inboard vortex interacts with the viscous free shear layer and diffuses, which explains why the pancake vortex is weaker in PIV than in the inviscid panel method prediction. Beyond the design condition, at $\alpha = 12°$, a conventional wingtip vortex reforms and coincides with the outboard displacement of the folded filament region. These results highlight the strong sensitivity of the Prandtl-D wake structure to angle of attack and support the benefits of the bell-shaped lift distribution near the design condition, where the wake remains effectively tip vortex free.

Apart from the empirical separation model applied on the wing surface, the panel method wake evolution is entirely inviscid. In this framework the vortex filaments fold over on themselves but do not merge with a viscous shear layer or experience core diffusion. Even with this limitation, the inviscid solution reproduces the main features of the measured wake, including the inboard folding at the design angle of attack and the disappearance and later re-emergence of the tip vortex. This agreement suggests that the global wake topology is governed primarily by inviscid processes, while the local vortex core strength and diffusion are controlled by viscous effects.

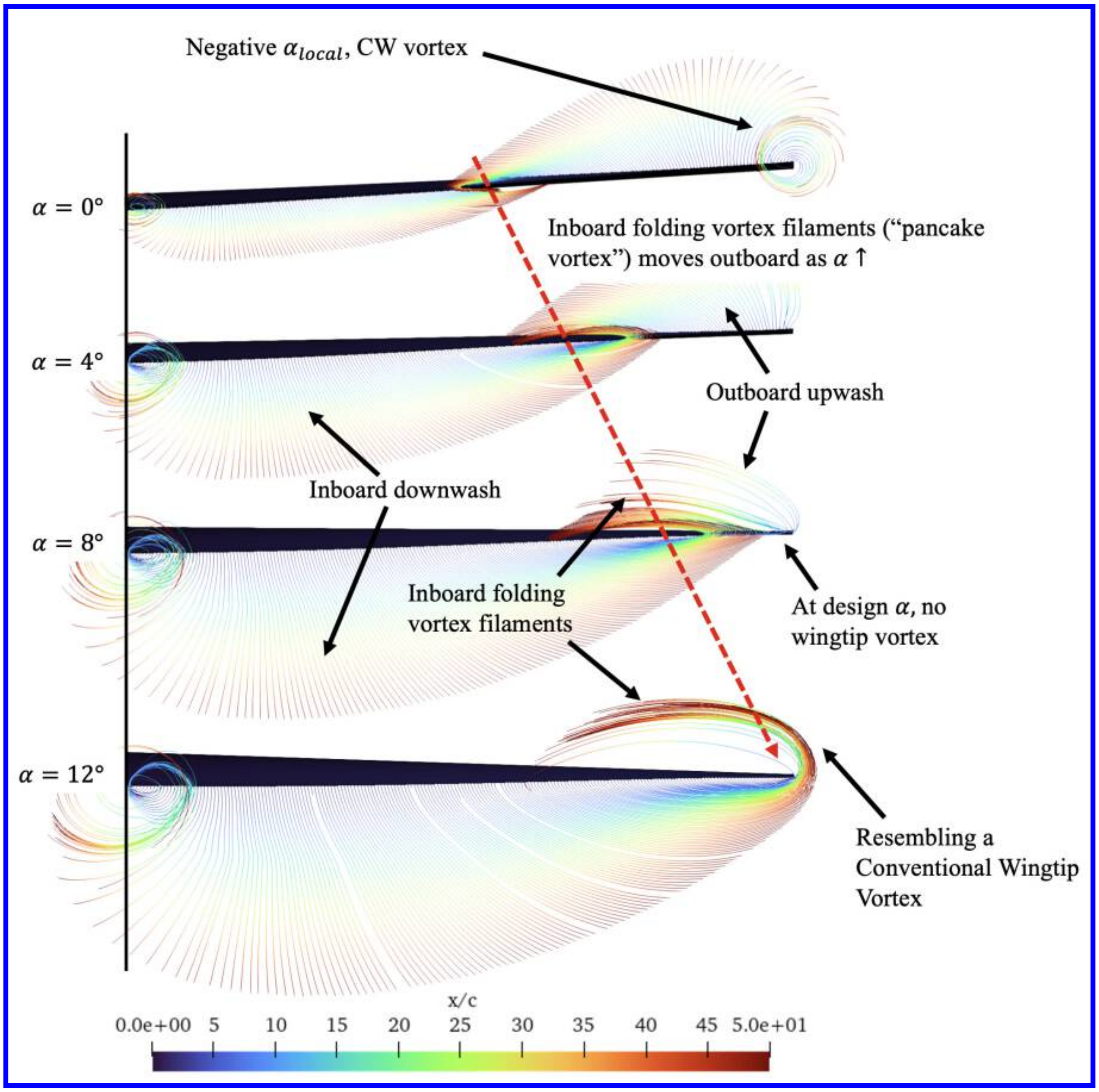

Figure 10. FlightStream Prandtl-D Vortex Filaments to $x/c = 50$

D. FlightStream Comparison to PIV

The FlightStream® results can be compared directly to PIV data. The nature of the inviscid FlightStream® panel method wake means the local properties in the wake will be different. But if the global behavior is the same it implies that behavior in the PIV data is driven by the similar mechanisms that are present in the FlightStream® model.

Figure 11 shows this comparison with both velocity and vorticity data. FlightStream® panel method cross-stream velocity and vorticity data are calculated based on discrete vortex filaments and their location. The results of this are evident in the Figure 11 results. Peaks of cross-stream velocity magnitude and vorticity coincide with the sparse filament locations in the wake. Still, the overall velocity vectors and shape of the wake align well with PIV results. Starting at α = 0°, the clockwise wingtip vortex and corresponding vorticity appears in both data frames, although the panel methods show discrete points of vorticity due to vortex filaments instead of it being continuous as seen in the PIV data. As seen before, as the angle of attack increases, the wingtip vortex weakens and then reverses direction. At the same time, an inboard area of circulating velocity and vorticity forms and moves outboard. By α = 12°, a conventional wingtip vortex has formed, although vorticity remains along the line of the inboard wake. One difference is that in the panel method case, at the design angle of attack, there is some increased peaks of the velocity and vorticity near the wingtip that are not observed in the PIV data. This may be the result of small angularity differences or a vortex filament structure that dissipates in viscous flow. Still, the global shape of the wake and patterns in the velocity vectors are consistent.

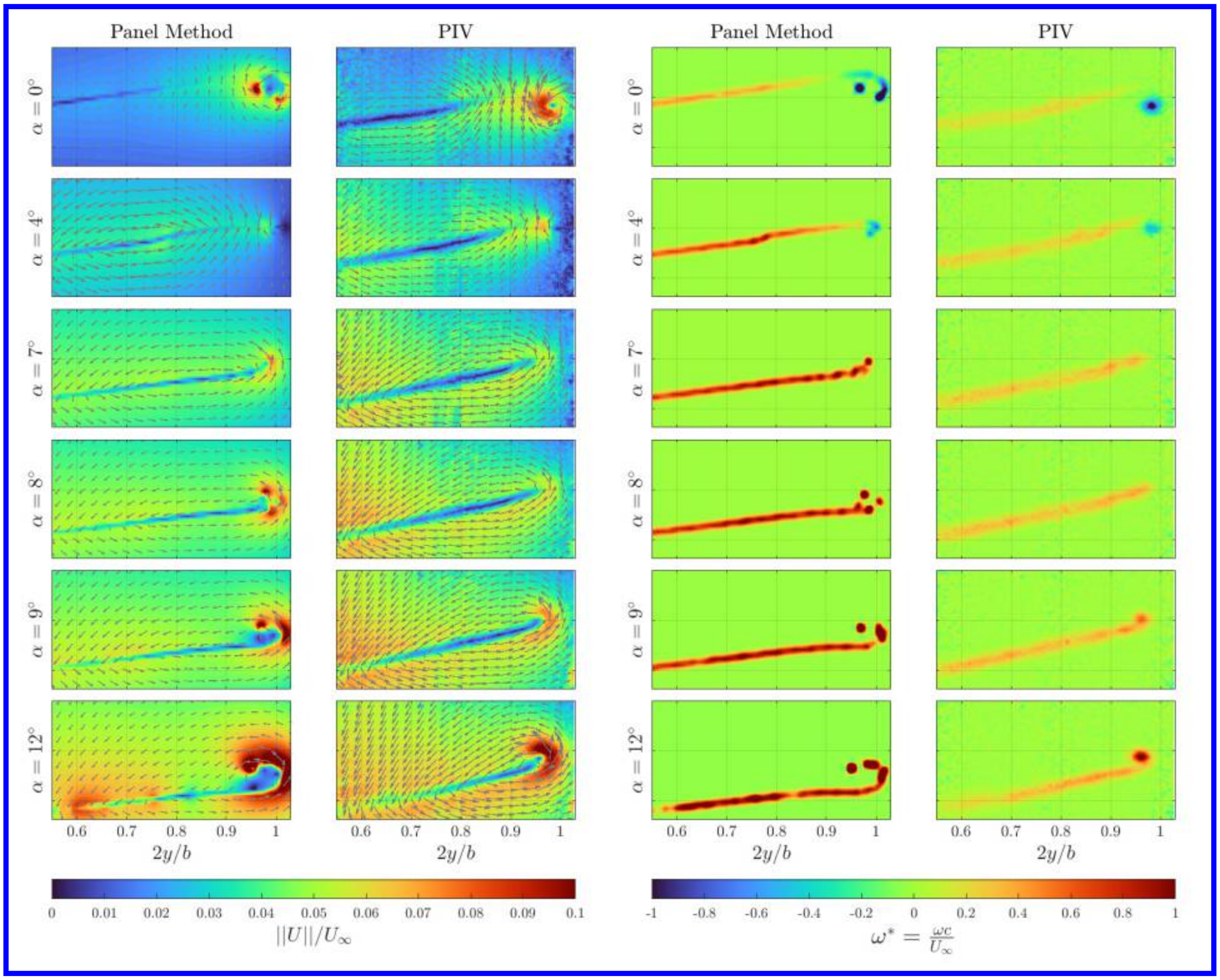

Figure 11. FlightStream (Panel Method) and PIV Velocity and Vorticity Comparison, 4c

IV. Conclusion

The results of this study provide experimental confirmation of the distinctive wake and vortex structure associated with the bell-shaped lift distribution and the Prandtl D wing. At the design angle of attack, $\alpha = 8°$, no conventional wingtip vortex is observed. Instead of concentrated vortices at the wingtips, the wake contains a broad inboard pancake vortex located at approximately $2y/b = 0.75$ to 0.78 and coincident with the wake shear layer. This flattened vorticity structure is substantially weaker and more diffuse than the concentrated vortex cores that appears on conventionally loaded wings.

Cross stream PIV shows that this wake topology is highly sensitive to angle of attack within a narrow operating range. At $\alpha = 0°$ and $\alpha = 4°$ the wake contains conventional wingtip vortices with concentrated vorticity and organized spiral patterns. At $\alpha = 7°$ the tip vortex is already strongly weakened, with circulation shifting inboard and peak vorticity decreasing. At the design condition, $\alpha = 8°$, the conventional tip vortex is fully suppressed. At $\alpha = 9°$ a tip vortex reappears and then strengthens with further increase in angle of attack. By $\alpha = 12°$ the wake is dominated by strong conventional tip vortices. This narrow two-degree window from $\alpha = 7°$ to $\alpha = 9°$ defines the range in which the wake remains tip vortex free, and it highlights the importance of precise angle of attack control in order to preserve the low vorticity wake. It is within these angles of attack where the maximum aerodynamic efficiency occurs as well. It also points to the need for future work that seeks to extend this favorable wake behavior over a wider lift range, for example through active lift distribution control.

The experimental findings show strong qualitative agreement with panel method simulations for all test conditions. The panel method is inviscid and represents the wake as persistent discrete vortex filaments, yet the overall shape of the wake and the velocity vector patterns match the PIV measurements. This agreement between an inviscid model and viscous experiments suggests that the global wake topology is set primarily by the spanwise circulation distribution of the bell-shaped loading, which can be predicted by inviscid methods, while viscous effects mainly govern the local vortex core strength and diffusion.

Taken together, the results provide clear experimental support for previous CFD predictions. Bell-shaped lift distributions on the Prandtl-D wing can suppress conventional wingtip vortex formation at design conditions and replace concentrated tip vortices with diffuse distributed vorticity embedded in the wake shear layer. The narrow angle of attack range associated with this behavior implies that practical implementations will benefit from control strategies that maintain the desired span loading across varying flight conditions, for example through active twist [25], distributed propulsion, or other mechanisms that modulate the lift distribution in flight.

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